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Department of Computer Science and Engineering Project Phase - I (BCS685)

Indian Market Portfolio Intelligence & Backtesting Platform

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Introduction

- Financial markets are highly dynamic, nonlinear, and influenced by economic, political, and investor sentiment factors.
- Traditional statistical methods often struggle to accurately predict stock price movements under volatile market conditions.
- Machine Learning and Deep Learning models such as LSTM, CNN, DNN, and Reinforcement Learning have shown improved forecasting and trading performance.
- Market regime detection techniques help classify markets into Bull, Bear, and Sideways phases for better strategy evaluation and risk management.
- This survey reviews recent research on stock market prediction, algorithmic trading, portfolio analytics, and regime-aware financial intelligence systems.

Literature Survey

S.No	Title of the Paper	Author(s)	Journal & Year	Methodology	Insights
1	Enhancing Stock Market Prediction: A Robust LSTM-DNN Model Analysis on 26 Real-Life Datasets	<i>K. Alam, M. H. Bhuiyan, I. U. Haque, M. F. Monir, T. Ahmed</i>	IEEE Access, 2024	Hybrid LSTM-DNN model using a sliding window approach for preprocessing	Improved prediction accuracy across 26 datasets; robust hybrid forecasting. Drawback: Highly susceptible to anomalies and data quality variations
2	Stock Price Prediction Using CNN and LSTM-Based Deep Learning Models	<i>S. Mehtab, J. Sen</i>	DASA, 2020	Univariate CNN and encoder-decoder LSTM architectures with walk-forward validation	High accuracy and fast execution for NIFTY 50 predictions. Drawback: Accuracy drastically decreases when significant market randomness occurs
3	An Application of Deep Reinforcement Learning to Algorithmic Trading	<i>T. Théate, D. Ernst</i>	Expert Systems with Applications, 2021	Trading Deep Q-Network (TDQN) using OHLCV data to maximize the Sharpe ratio	Outperforms benchmark strategies and displays robustness to trading costs. Drawback: High variance limits generalizability due to the bias-variance dilemma
4	Machine Learning Based Stress Testing Framework for Indian Financial Market Portfolios	<i>V. Sagar G, S. Ali, S. P. Chakrabarty</i>	arXiv Preprint, 2025	Rolling-window pipeline extracting latent risk factors via PCA, AE, and VAE	Effectively models linear and nonlinear sectoral dependencies during stress scenarios. Drawback: Focuses only on equity indexes, ignoring sentiment and dynamic hedging constraints

Literature Survey

S.No	Title of the Paper	Author(s)	Journal & Year	Methodology	Insights
5	A Hybrid Learning Approach to Detecting Regime Switches in Financial Markets	<i>P. Akiyamen, Y. Z. Tang, H. Hussien</i>	ICAIF, 2020	PCA for dimensionality reduction combined with k-means clustering and supervised classification	Generated positive alpha and reduced drawdowns for tail-hedging strategies. Drawback: Differenced time-series inputs inherently lack long-term memory
6	Regime-Switching Factor Investing with Hidden Markov Models	<i>M. Wang, Yi-Hong Lin, I. Mikhelson</i>	Journal of Risk and Financial Management, 2020	Gaussian Hidden Markov Model (HMM) to classify broad market regimes dynamically	Delivered higher annualized returns and Sharpe ratios than static factor models. Drawback: Performance remains limited by the alpha-generation capability of underlying factor models
7	A Multi-Market Data-Driven Stock Price Prediction System: Model Optimization and Empirical Study	<i>L. Yu, Y. Chen, Y. Zheng</i>	IEEE Access, 2025	GRU-based deep learning trained on a cross-national, multi-asset dataset	GRU significantly outperformed baseline models in daily returns and accuracy. Drawback: Limited generalizability as it was tested solely on one Chinese index
8	Stock Market Prediction With Transductive Long Short-Term Memory and Social Media Sentiment Analysis	<i>Ali Peivandizadeh, et al.</i>	IEEE Access, 2024	Integrates Off-policy PPO for sentiment imbalance and Transductive LSTM (TLSTM) for temporal dynamics	Significant reduction in prediction errors and improved minority sentiment classification. Drawback: High computational complexity and overfitting risks during extreme market volatility

Existing Platforms Survey

S.No	Platform Name	Category	Key Strengths / Features	Limitations
1	Groww, INDmoney, Moneycontrol	<i>Retail Tracking Apps</i>	Excellent for portfolio tracking, basic XIRR, and fundamental news	Completely lacks strategy backtesting, market regime awareness, and risk-adjusted metrics
2	Zerodha Streak	<i>Rule-Based Trading</i>	Offers basic indicator-based strategy building and automated execution	Restricted to simple technical rules; weak ML evaluation and no market regime intelligence
3	Smallcase	<i>Thematic Investment</i>	Provides curated thematic stock baskets for portfolio-level investing	No dynamic regime adaptation or automated strategy adjustments based on market conditions
4	TradingView	<i>Advanced Global Charting</i>	Powerful visualization, deep charting tools, and manual historical backtesting	Lacks built-in Indian market regime context and adaptive ML-driven strategy recommendations
5	QuantConnect	<i>Quantitative Infrastructure</i>	Deep multi-asset historical backtesting and advanced algorithmic execution	Highly complex and requires coding knowledge, making it completely inaccessible to average retail investors

Problem Identification

Blind Strategy Use

Retail investors apply RSI, MA, and momentum indicators without any historical validation or risk assessment.

No Regime Awareness

Strategies are evaluated as if markets are uniform. Bull, bear, and sideways regimes have fundamentally different risk-return dynamics.

Fragmented Analytics

Existing tools compartmentalize prediction, backtesting, and risk into separate workflows — no integrated decision support.

Backtest Overfitting

Standard backtesting lacks proper walk-forward validation, leading to overconfidence in historical results that fail in live markets.

Accessibility Gap

Advanced quant tools (QuantConnect, Backtrader) require coding expertise. No Indian-market-focused platform serves non-technical retail users.

Risk Blind Spots

Most platforms show returns without quantifying downside exposure — Sharpe, Max Drawdown, and volatility bands are absent.

Problem Statement

There is no structured, accessible platform that simultaneously backtests investment strategies on Indian equity markets, detects prevailing market regimes, and provides ML-driven risk-aware strategy recommendations — leaving retail investors without a validated, interpretable decision-support system.

The system must be able to answer:

1. How did the strategy perform historically
2. Under what regime does it work best?
3. Will it likely work in current conditions?
4. What is the downside risk if used now?

Scope

IN SCOPE

- NIFTY 50 index historical OHLCV data via yfinance
- 6 predefined strategies: Buy & Hold, Moving Average Crossover, RSI, Momentum, Bollinger Bands, and Dual Momentum
- Market regime detection using K-Means clustering (Bull / Bear / Sideways)
- Performance metrics: CAGR, Sharpe Ratio, Max Drawdown, Volatility
- ML strategy recommendation via Random Forest classifier
- Probabilistic risk forecasting using bootstrap resampling
- Dual interactive dashboards (Beginner and Professional) for deep analytics

OUT OF SCOPE

- Real-time live trading or order execution
- Exact deterministic future price prediction
- User-defined custom strategies
- Options, futures, or derivatives analysis
- Real-time streaming data pipelines
- Multi-stock portfolio-level optimization (Markowitz - future work)

Goals and Objectives

- To study and analyze recent Machine Learning and Deep Learning techniques used in stock market prediction and trading systems.
- To examine market regime detection approaches such as Hidden Markov Models, clustering, and regime-switching frameworks.
- To identify limitations in existing financial analytics platforms and ML-based trading models.
- To explore methods for improving portfolio analysis, risk management, and strategy evaluation using regime-aware systems.
- To propose a conceptual framework for an AI-driven market intelligence and portfolio analytics system tailored for Indian equity markets.

Methodology

Observe

Ingest real-world OHLCV data from the NIFTY 50.

Classify

Apply Unsupervised Learning (K-Means) to identify the market's current structural state.



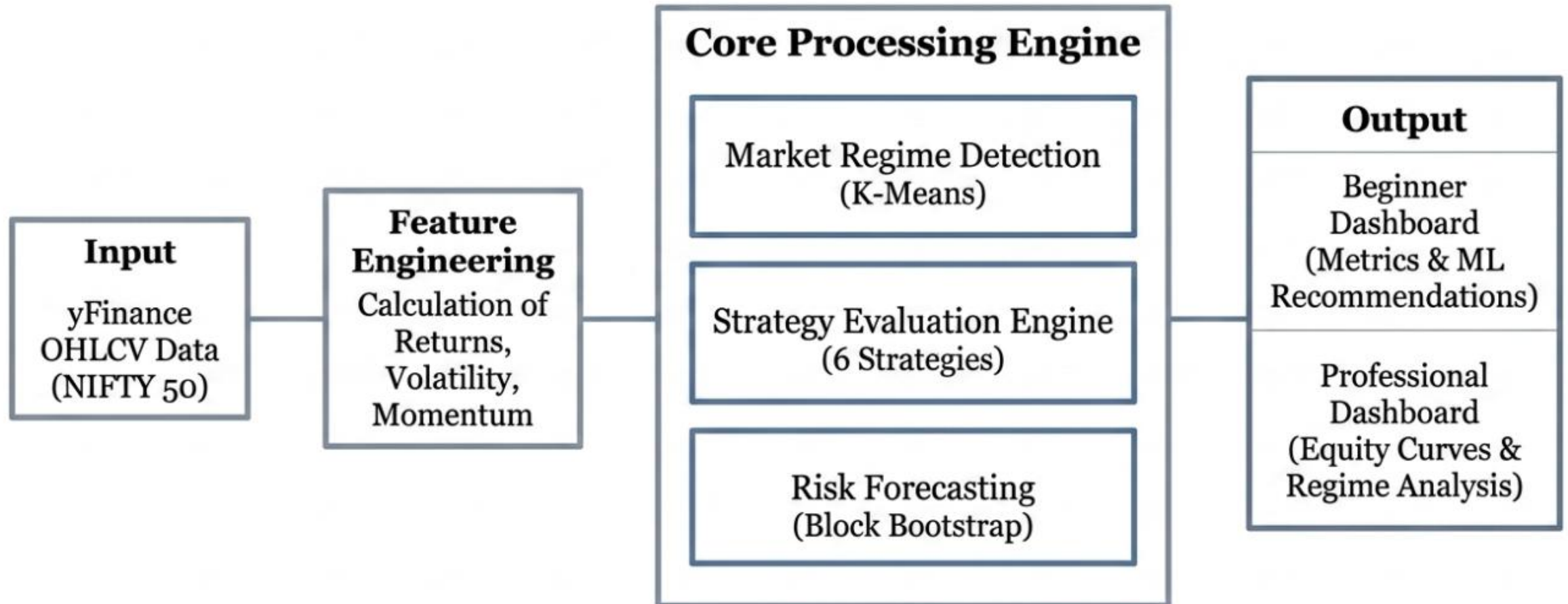
Recommend

Apply Supervised Learning (Random Forest) to output the highest-probability strategy action, abandoning exact price targets.

Simulate

Run historical backtests across the 6 quantitative strategies within that specific mathematical state.

Data Flow



Contribution to the Society

- Helps retail investors make more informed and data-driven investment decisions using AI-based market analysis.
- Improves awareness of market risk by identifying changing market regimes and volatility conditions.
- Encourages the use of Machine Learning and financial analytics in the Indian equity market ecosystem.
- Provides a unified framework for portfolio evaluation, strategy analysis, and risk forecasting in a simplified manner.
- Supports students, researchers, and beginner investors in understanding practical applications of AI in finance and algorithmic trading.

References

Journal Papers:

- [1] K. Alam, M. H. Bhuiyan, I. U. Haque, M. F. Monir, and T. Ahmed, "Enhancing Stock Market Prediction: A Robust LSTM-DNN Model Analysis on 26 Real-Life Datasets", IEEE Access, Vol-12, pp. 122757–122768, 2024.
- [2] M. Wang, Y.-H. Lin, and I. Mikhelson, "Regime-Switching Factor Investing with Hidden Markov Models", Journal of Risk and Financial Management, Vol-13, p. 311, 2020.
- [3] T. Théate and D. Ernst, "An Application of Deep Reinforcement Learning to Algorithmic Trading", Expert Systems with Applications, Vol-173, p. 114632, 2021.
- [4] H. Arian, D. Norouzi Mobarekeh, and L. Seco, "Backtest Overfitting in the Machine Learning Era: A Comparison of Out-of-Sample Testing Methods in a Synthetic Controlled Environment", Knowledge-Based Systems, Vol-305, p. 112477, 2024.

Conference Papers:

- [5] P. Akiyamen, Y. Z. Tang, and H. Hussien, "A Hybrid Learning Approach to Detecting Regime Switches in Financial Markets", ACM International Conference on AI in Finance (ICAIF '20) in New York, NY, USA, October 15th and 16th 2020.

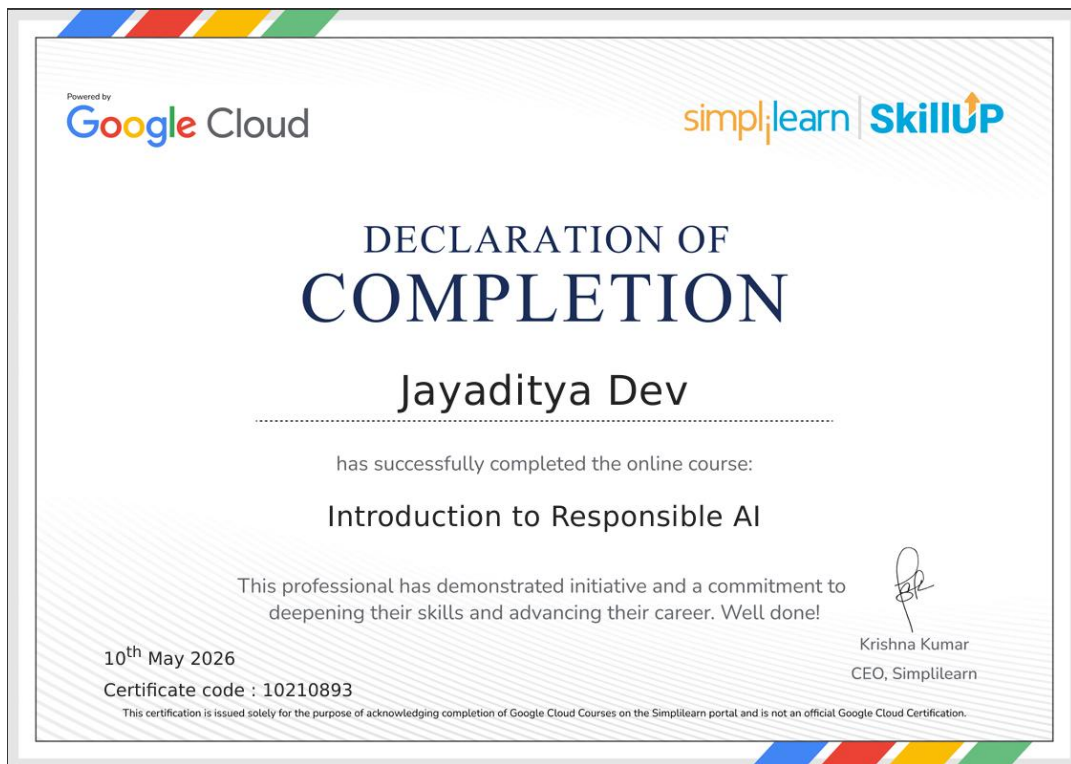
Website: [6] <http://finance.yahoo.com>, Accessed on 01-05-2026

Published Survey Paper

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